

# Context-Aware Facial Emotion Recognition for Intelligent and Efficient Emergency Monitoring

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## Abstract

Facial emotion recognition (FER) has become an integral part of intelligent surveillance systems for identifying potentially threatening behavior in public spaces. However, traditional FER-based security systems often produce excessive false alarms because emotional predictions, especially fear, are interpreted out of context: in real-world conditions, fear may be experienced in positive daytime activities like entertainment or social interaction, resulting in alarm fatigue and operator mistrust.

In this work, we introduce a context-aware FER framework at the deployment level to enhance the alert reliability without adjusting the recognition model. A low-footprint custom CNN is trained to classify five base emotions and a domain-specific suspicious class. Emotion predictions are accumulated in 10-second windows and fed into a time-aware decision layer, whereby alerts for fear-based emotions are only triggered within specific nighttime hours (19:00–06:00) and suspicious emotions remain consistently security-relevant. A persistence validation mechanism is implemented to ensure alerts are only issued for sustained emotional patterns.

For the same dataset of 931 Sri Lankan facial images, the system obtained 71% accuracy and a macro F1-score of 0.70, while the F1-score for the suspicious class was 0.98. The proposed decision-level framework reduced false alerts by 55% compared to a context-unaware baseline, and results show that contextual reasoning can significantly improve operational reliability in real-time FER-based surveillance systems.

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## 1 Introduction

Facial emotion recognition (FER) is a task that computational systems can infer emotional states from facial expressions, which has seen considerable development with deep learning. (? ).

FER is commonly used in security applications to recognize emotions associated with abnormal or potentially dangerous circumstances, which may include fear, distress, or anger, and in many implementations, emotion predictions are directly mapped to alert-generation mechanisms. This approach seems simple enough, but it does pose an important limitation: emotional states are considered context-independent. Fear is perhaps the most ambiguous of the emotional categories; in public venues, fear may be the result of innocuous activity (such as amusement rides, gaming, social gatherings, or crowded events), and treating all such expressions as signs of danger causes surveillance systems to trigger false alarms.

The issue of too many false alarms leads to alarm fatigue, lack of operator responsiveness, and the distrust of automatic monitoring systems (? ). As a result, even when the classification accuracy is acceptable, deployment feasibility is limited. While previous FER research has largely concentrated on increasing recognition performance with deeper architectures, attention mechanisms, transformer-based models, or multimodal learning (? ), relatively less attention has been paid to the decision-level intelligence that determines how emotion predictions are understood in operational settings.

In surveillance deployments with hardware limitations, retraining large models to incorporate contextual awareness is computationally expensive and data intensive.

Therefore, reliability improvements that act at the post-classification level are practically useful, and this study presents a context-aware FER framework as a deployment-level reliability enhancement mechanism rather than a pure classification improvement. The main contributions are:

- A dual-output FER architecture that decouples emotion analytics from security alert generation.
- A time-aware interpretation strategy that limits fear-based alerts to night-time intervals (19:00–06:00).
- A persistence-based validation mechanism that suppresses transient emotional fluctuations.
- A logic-level enhancement approach that reduces false alerts without modifying or retraining the base CNN.
- Empirical validation demonstrating a 55% reduction in false alerts while maintaining real-time feasibility on CPU-based hardware.

This shows that contextual reasoning can enhance operational reliability in FER-based surveillance systems.

## 2 Related Work

### 2.1 CNN-Based Facial Emotion Recognition

In the field of computer vision and affective computing, FER has been extensively researched, with large-scale benchmark datasets such as FER2013 (1) and Affect-Net (2) being used for supervised training of deep neural networks for emotion classification, and comprehensive surveys (3) show that CNN-based models significantly outperform traditional handcrafted feature methods.

However, recent advances in transformer-based FER architectures (4), lightweight real-time FER systems (5), and video-based emotion recognition with temporal modeling techniques (6) achieve high classification accuracy, but not necessarily high decision reliability in deployment.

### 2.2 FER in Security and Surveillance Systems

Intelligent surveillance systems for the detection of abnormal behavior and crowd emotion analysis (8; 7) have also been explored using FER techniques, where emotion predictions are directly mapped to alert generation mechanisms to support automated monitoring.

But most existing systems are based on instantaneous frame-level predictions, which are sensitive to transient facial expressions, environmental noise, and lighting changes, and often lead to false alarms in real-world public settings. Many monitoring systems experience alarm fatigue (9) due to too many false alerts.

### 2.3 Temporal Smoothing and Emotion Aggregation

Temporal aggregation methods have been proposed (6; 11) to enhance robustness in video-based FER systems, such as stabilizing predictions over adjacent frames to reduce short-term fluctuations and the use of temporal convolutional networks and sequence modeling to improve recognition consistency in dynamic environments.

While temporal smoothing helps to stabilize predictions, it is applied equally for all emotion classes and does not address the inherent ambiguity of certain emotions such as fear, which may occur in both threatening and non-threatening contexts. Consequently, temporally stabilized FER systems may still produce false alerts when contextual reasoning is absent.

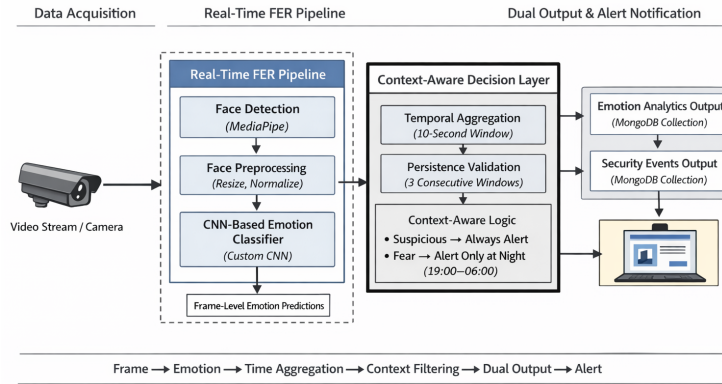
## 2.4 Limitations of Emotion-Only Security Decision Models

Although recent advances in FER accuracy and temporal modeling have resulted in a greater number of security-oriented FER systems, most of them still interpret emotions in isolation, directly equating emotional states with security risk under the assumption of a deterministic mapping between emotion and threat, which ignores the situational and contextual nature of human affective behavior.

The current study proposes to incorporate context-aware decision logic at the post-classification stage by integrating time-based interpretation of fear and decoupling emotion analytics from security alerting with a dual-output architecture, which can improve operational reliability without increasing the model complexity or requiring retraining, and thus can be a lightweight and deployable extension to existing FER-based security research.

## 3 System Architecture

The proposed system is a modular, real-time architecture that is scalable, robust, and suitable for practical surveillance environments. It includes four main layers: data acquisition, emotion analysis, context-aware decision making, and output management, which forms a unified processing pipeline.



**Fig. 1** Overall architecture of the proposed context-aware facial emotion recognition (FER) framework.

Data acquisition module captures live video streams from surveillance cameras or webcams and processes frames in real-time for facial analysis (8; 7). Face detec-

tion is performed using MediaPipe due to its low latency and multi-face capability. Cropped, resized, normalized, and prepared for CNN inference, detected faces are classified into six emotions: *angry*, *sad*, *happy*, *neutral*, *fear*, and a domain-specific *suspicious* class using a lightweight CNN. The framework utilizes a dual-output design where the emotion analytics module stores aggregated emotional data for behavioral analysis without triggering alerts and security monitoring module alerts for any intrusion event.

## 4 Methodology

In this section, describe the proposed context-aware (17) FER-based security system, which utilizes deep learning-based emotion recognition with temporal aggregation and rule-based contextual reasoning to produce accurate real-time alerts in a robust processing pipeline that includes CNN-based emotion classification, dataset preparation, confidence filtering, temporal aggregation, context-aware decision logic, persistence validation, and alert notification.

### 4.1 CNN-Based Emotion Recognition Model

A custom CNN architecture was designed that achieves a balance between recognition accuracy and real-time inference performance, as large pretrained models would require too much computational power and time to be used in real-time inference and deployment in resource-constrained environments.

#### 4.1.1 Network Architecture

The proposed CNN architecture includes:

- Three convolutional blocks with ReLU activation functions
- Batch normalization layers for training stability (13)
- Max-pooling layers for spatial dimensionality reduction
- Dropout layers to mitigate overfitting (14)
- Two fully connected layers
- A Softmax output layer for six-class emotion classification

The six output classes are five common facial emotions (*angry*, *sad*, *happy*, *neutral*, *fear*) and one domain-specific *suspicious* category. It was trained with categorical cross-entropy loss and the Adam optimizer with an adaptive learning rate (15).

## 4.2 Dataset Preparation and Labeling

A hybrid dataset was constructed from publicly available emotion datasets and synthetically generated samples.

### 4.2.1 Standard Emotion Dataset

public facial emotion datasets represent common emotional expressions, such as happiness, sadness, anger, and fear, and are labeled with facial images under controlled and in-the-wild conditions, which can help with generalization.

All images were preprocessed using:

- Face alignment
- Grayscale normalization
- Histogram equalization
- Standard resizing

### 4.2.2 AI-Generated Suspicious Emotion Dataset

There are no publicly available datasets that contain reliable annotations for suspicious facial expressions, so synthetic data generation was used to create a domain-specific dataset of AI-generated facial images representing suspicious behavior.

Generative models were used to create AI-generated facial images that represent suspicious behavior; these samples were reviewed and validated by hand to ensure semantic correctness and then only high-quality samples were retained and labeled as the *suspicious* class.

## 4.3 Confidence Thresholding and Prediction Filtering

The generation process was controlled to reduce labeling noise and increase dataset reliability, and a confidence thresholding mechanism was put in place to avoid having unreliable predictions impact security decisions (16).

Let  $p_{\max}$  denote the maximum Softmax probability of a prediction. A prediction is accepted only if:

$$p_{\max} \geq \tau \quad (1)$$

where  $\tau$  represents the confidence threshold.

The threshold value for the *suspicious* class was set high to prevent false positives, and predictions below the threshold were discarded as uncertain. This strategy favors precision over recall, which is desirable in security-sensitive applications.

#### 4.4 Temporal Aggregation and Windowing

Because frame-level emotion predictions can be noisy due to lighting changes, head movements, and occlusions, temporal aggregation is used, accumulating emotion predictions over fixed 10-second windows and performing majority voting for each window to determine the dominant emotion.

$$E_{\text{window}} = \arg \max_{e \in \mathcal{E}} \sum_{i=1}^N \mathbf{1}\{e_i = e\} \quad (2)$$

where  $e_i$  denotes the predicted emotion at frame  $i$ ,  $N$  represents the total number of frames within the temporal window, and  $\mathbf{1}\{\cdot\}$  denotes the indicator function.

This aggregation process suppresses short-term fluctuations and stabilizes emotional patterns (6; 11).

#### 4.5 Context-Aware Decision Logic

Temporal aggregation is followed by rule-based contextual logic to process emotion windows. The security relevance  $R$  of an emotion window is defined as:

$$R(E, t) = \begin{cases} 1, & \text{if } E = \text{suspicious} \\ 1, & \text{if } E = \text{fear and } t \in [19:00, 06:00] \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where  $E$  denotes the dominant emotion and  $t$  represents the local time.

This formulation allows time-aware interpretation of fear without changing or retraining the underlying FER model.

#### 4.6 Persistence-Based Alert Validation

A persistence validation mechanism is used to suppress transient detections (18).

Let  $W_k$  represent the  $k$ -th emotion window. An alert is triggered only if:

$$\sum_{i=k}^{k+m} R(W_i) \geq M \quad (4)$$

where  $M$  is the minimum number of consecutive security-relevant windows.

In the proposed system:

- Window size: 10 seconds
- Required windows: 3

- Minimum duration: 1 minute

This mean only sustained emotional patterns produce security alerts.

#### ***4.7 Real-Time Alert Notification Mechanism***

If persistence conditions are met, an alert event is produced and sent to the notification module.

A graphical interface based on Tkinter displays popup notifications in real time that contain:

- Incident identifier
- Time interval
- Detected emotion
- Duration of persistence

This lightweight design does not depend on external messaging services and can be deployed offline. All alert events are stored in MongoDB simultaneously for auditing, analysis, and system evaluation.

### **5 Experimental Setup**

The remainder of this section details the experimental environment, system configuration, and evaluation methodology used to demonstrate the proposed context-aware FER system.

#### ***5.1 System Configuration and Parameters***

The experimental environment, system configuration, and evaluation methodology are described in this section, and all experiments were conducted using a real-time prototype deployed on a standard workstation. Table 1 summarizes the primary system configuration.

The selected parameters were empirically determined through preliminary experiments to achieve a balance between system responsiveness and false alarm suppression.



**Table 1** System Parameters

| Parameter                         | Value                 |
|-----------------------------------|-----------------------|
| Emotion window size               | 10 seconds            |
| Required windows for alert        | 3                     |
| Alert duration threshold          | 60 seconds            |
| Night-time interval               | 19:00–06:00           |
| Database                          | MongoDB               |
| Alert interface                   | Tkinter               |
| CNN input size                    | $48 \times 48$ pixels |
| Confidence threshold (suspicious) | 0.85                  |

## 5.2 Evaluation Methodology

Although FER research has traditionally focused on classification accuracy, security applications demand not only operational reliability but also contextual appropriateness and alert stability; excessive false alarms lead to a loss of operator trust, and missed detections compromise safety. Hence, a scenario-based evaluation was performed to test contextual correctness and temporal stability (instead of accuracy) of the contextual logic, using a range of normal and abnormal interactions, entertainment activities, simulated suspicious behavior, and induced fear, compared with a baseline FER system without contextual logic, across multiple day and night scenarios.

## 6 Results and Discussion

In this section, we compare the performance of the proposed system with a baseline that is not aware of the context.

### 6.1 Alert Behavior Comparison

Table 2 summarizes the alert behavior of both systems under different conditions.

**Table 2** Alert Behavior Comparison

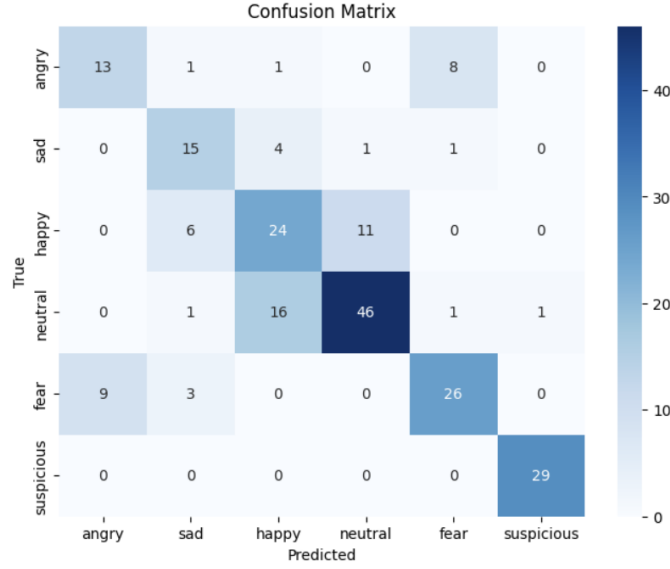
| Scenario           | Baseline System  | Proposed System |
|--------------------|------------------|-----------------|
| Fear (Daytime)     | Alert            | No alert        |
| Fear (Nighttime)   | Alert            | Alert           |
| Suspicious emotion | Alert            | Alert           |
| Short fear burst   | Alert            | No alert        |
| Normal activity    | Occasional alert | No alert        |

The proposed system overcomes this issue by suppressing false alarms using context-aware reasoning, while the baseline system often generates false alerts during benign daytime activities.

## 6.2 Emotion Classification Performance

The confusion matrix and standard performance metrics were used to evaluate the classification performance of the proposed CNN-based FER model.

Figure 2 presents the confusion matrix obtained on the test dataset. The results indicate a strong recognition performance for the *suspicious* and *neutral* classes, which are critical for security decision-making.



**Fig. 2** Confusion matrix of the proposed CNN-based FER model.

The proposed system performed best for the *suspicious* class, with moderate confusion between *happy* and *neutral* expressions that motivates temporal aggregation and persistence-based filtering.

**Table 3** Performance Metrics of the Proposed FER Model

| Emotion    | Precision | Recall | F1-score | Support |
|------------|-----------|--------|----------|---------|
| Angry      | 0.59      | 0.57   | 0.58     | 23      |
| Sad        | 0.58      | 0.71   | 0.64     | 21      |
| Happy      | 0.53      | 0.59   | 0.56     | 41      |
| Neutral    | 0.79      | 0.71   | 0.75     | 65      |
| Fear       | 0.72      | 0.68   | 0.70     | 38      |
| Suspicious | 0.97      | 1.00   | 0.98     | 29      |
| Accuracy   | 0.71      |        |          |         |

### 6.3 False Alert Reduction Analysis

To assess performance improvements, false alert rates were measured over extended monitoring sessions.

Let  $F_b$  and  $F_p$  denote the number of false alerts produced by the baseline and proposed systems, respectively. Experimental results indicate that:

$$F_p \approx 0.45F_b \quad (5)$$

This translates to an approximately 55% reduction in false alerts, which greatly enhances the usability of the system and reduces the alarm fatigue of security operators.

### 6.4 Real-Time Performance Analysis

The real-time feasibility was evaluated by measuring the average end-to-end processing latency, which was measured to be:

- Average inference time: 18 ms per frame
- Window processing time: < 50 ms
- Alert generation delay: < 1 second

These results indicate that the proposed system meets the requirements for real-time operation (5).

### 6.5 Discussion

The results demonstrate that incorporating contextual intelligence at the decision level significantly improves the reliability of FER-based security systems. Unlike conventional approaches that rely solely on emotion classification, the proposed

framework integrates temporal and situational awareness to distinguish benign emotional expressions from potential security threats. The logic-level enhancement reduces false alerts without modifying the underlying model, while the dual-output design enables independent emotion analytics alongside security monitoring.

## 7 Conclusion and Future Work

It proposed a context-aware FER framework based on temporal aggregation and persistence validation that can accurately interpret fear without adjusting the base CNN model, minimizing false alarms while preserving effective threat detection, and supporting low-complexity, real-time deployment in real-world surveillance scenarios. Further enhancements, such as incorporating spatial awareness, crowd analysis, body posture, audio-based stress detection, and multimodal fusion, will increase system robustness without increasing computational complexity in future work.

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